Project – Interactive Dashboard using Pivot Tables and Charts

In this project, I created an interactive dashboard in Excel using pivot tables, pivot charts, and slicers to visualize multi-year sales data. The goal was to build a dynamic report that updates seamlessly with new data and allows filtering across multiple views.

1. I formatted the sales dataset (2017–2019) in **Data Sheet** as an **Excel Table** using Ctrl + T, ensuring structured referencing and future-proof data expansion.

A	B	C	D	E
Year	Category	Product	Sales	Rating
2017	Accessories	Pumps	$ 700	10%
2017	Accessories	Helmets	$ 8,300	99%
2017	Accessories	Tires and Tubes	$ 8,700	90%
2017	Accessories	Locks	$ 10,000	85%
2017	Accessories	Bike Racks	$ 300	5%
2017	Accessories	Lights	$ 1,300	90%
2017	Bikes	Road Bikes	$ 3,500	50%
2017	Bikes	Mountain Bikes	$ 3,100	35%
2017	Bikes	Touring Bikes	$ 500	22%
2017	Bikes	Cargo Bike	$ 3,200	48%
2017	Clothing	Socks	$ 3,700	22%
2017	Clothing	Shorts	$ 12,200	56%

2. I created 5 pivot tables in another Sheet named as **Working**, which I used later to create charts.

Sum of Sales	Column Labels				
Row Labels	Accessories	Bikes	Clothing	Components	Grand Total
2017	$29,300	$10,300	$44,800	$26,700	$111,100
2018	$120,700	$23,100	$60,400	$45,100	$249,300
2019	$233,800	$37,800	$104,200	$62,100	$437,900
2020	$235,956	$36,619	$121,161	$70,176	$463,912
Grand Total	$619,756	$107,819	$330,561	$204,076	$1,262,212

Sum of Sales	Column Labels			
Row Labels	Accessories	Bikes	Clothing	Components
2017				
2018	312%	124%	35%	69%
2019	94%	64%	73%	38%
2020	1%	-3%	16%	13%

Average of Rat	Column Labels				
Row Labels	Accessories	Bikes	Clothing	Components	Grand Total
2017	63%	39%	35%	50%	46%
2018	88%	37%	48%	42%	54%
2019	94%	48%	56%	50%	62%
2020	90%	67%	51%	34%	58%
Grand Total	84%	48%	48%	44%	55%

Average of Rat	Column Labels			
Row Labels	Accessories	Bikes	Clothing	Components
2017				
2018	39%	-5%	40%	-14%
2019	7%	31%	16%	18%
2020	-4%	38%	-9%	-31%

Row Labels	Sum of Sales
Accessories	619,756
Tires and	141,810
Locks	107,875
Helmets	97,227
Lights	94,282
Bike Racks	93,692
Pumps	84,870
Clothing	330,561

PivotTable Fields
Choose fields to add to report:
Drag fields between areas below:
Search
Year
Category
Product
Sales
Rating
More Tables...
Filters
Columns: Category
Rows: Year
Values: Sum of Sales
Defer Layo... Update

Dashboard | Workings | Data | New Data

3. I inserted the first **pivot chart** to visualize sales by product category and added a **Year slicer** for filtering. I formatted the chart as a **bar chart**, applied number formatting with comma separators, added data labels, and removed unnecessary elements (e.g., gridlines, axis).
4. I created a **line chart** to display sales over time by category and connected it to a separate **Category slicer**. The slicer was formatted with four columns and customized for a clean appearance.
5. A third **line chart** was added to show average ratings over time. This involved changing the pivot table's value field settings to calculate **average** and applying **percentage formatting**.
6. To analyze **year-on-year percentage changes**, I created two pivot tables and applied **“Show Values As > % Difference From > Previous Year”** for both sales and ratings. The charts were inserted as **column charts** with axis formatting adjusted for potential negative values.

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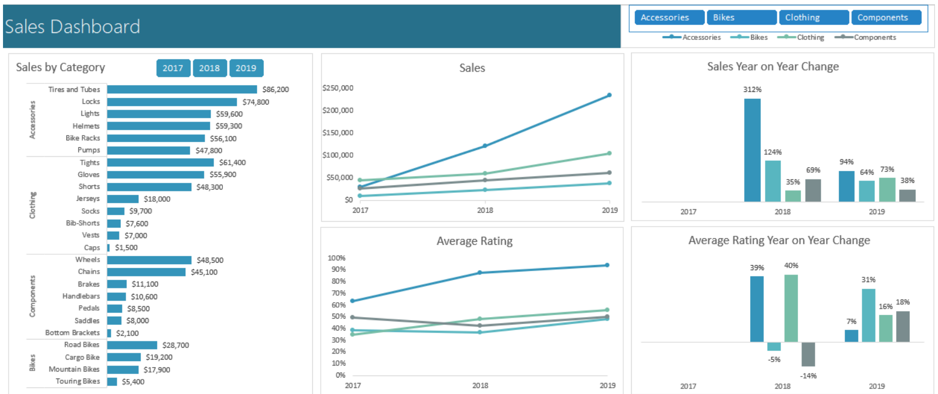
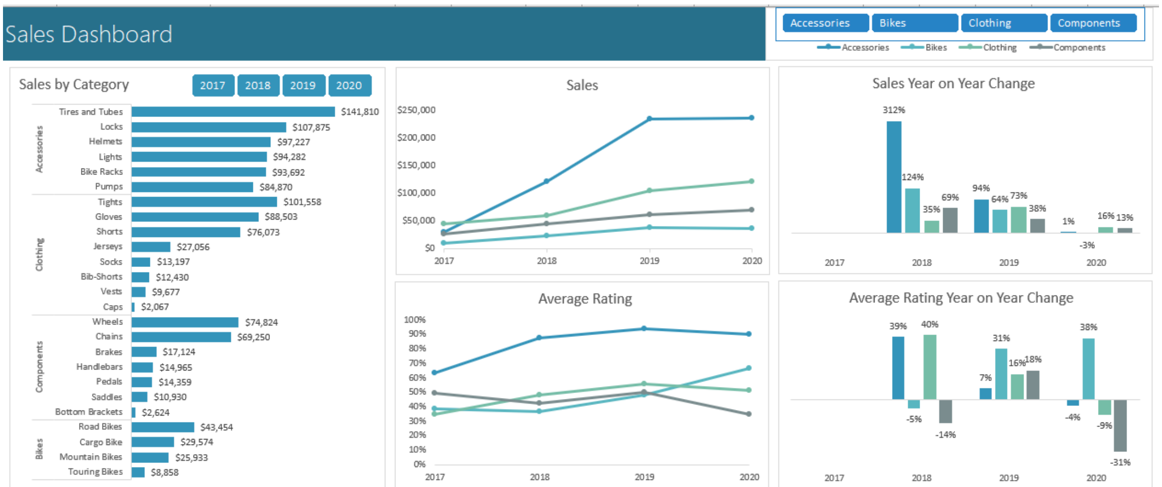
7. All pivot charts and slicers were assembled on the **Dashboard** sheet using precise alignment via the **Alt** key for grid snapping. **Report Connections** were configured for each slicer to ensure correct filtering behavior.



8. I added a **dummy chart** to serve as a **shared legend**, minimizing chart clutter. The legend chart was aligned behind the slicers on the dashboard.
9. Finally, I tested the dashboard's dynamic capability by appending 2020 data to the source table and clicking **Refresh All**, which successfully updated all visuals instantly.



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